HDOC-51

Next Greater Element Using Pointers and Nested Loops

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named **next_greater()** using its address. Use pointers to pass the array between the calling function and next_greater(). The function must find the nearest greater element to the right of every array element, display -1 if no greater element exists, print results in array order separated by commas, and use nested loops.

Aim

To implement a C program using pointers and nested loops to find the next greater element to the right for every element of a variable-length integer array.

Algorithm

1. Start the program.

2. Read the size n of the array and declare the variable-length array arr[n].

3. Read n integer elements into arr.

4. Pass the address of arr to the function next_greater(arr, n).

5. For each element arr[i], initialize next to -1.

6. Using an inner loop, scan elements from index i + 1 toward the right.

7. When the first element greater than arr[i] is found, store it in next and break the inner loop.

8. Print next for the current element. If none is found, -1 remains the result.

9. Print commas between results and finish after all elements are processed.

10. Stop.

C Program

```
#include <stdio.h>
```

```
void next_greater(int *arr, int n)
{
   int i, j, next;

   for (i = 0; i < n; i++)
   {
      next = -1;

      for (j = i + 1; j < n; j++)
      {
         if (*(arr + j) > *(arr + i))
         {
            next = *(arr + j);
            break;
         }
      }

      printf("%d", next);
      if (i < n - 1)
         printf(", ");
   }
}

int main(void)
{
   int n, i;

   printf("Enter the size of the array: ");
   scanf("%d", &n);

   if (n <= 0)
   {
      printf("Array size must be positive.\n");
      return 0;
   }

   int arr[n];

   printf("Enter %d elements: ", n);
   for (i = 0; i < n; i++)
      scanf("%d", &arr[i]);

   printf("Next greater elements: ");
   next_greater(arr, n);
   printf("\n");

   return 0;
```

```
}
```

Test Cases

Test	n	Array elements	Expected output	Purpose
1	5	4 5 2 10 8	5, 10, 10, -1, -1	Mixed values
2	5	1 2 3 4 5	2, 3, 4, 5, -1	Strictly increasing
3	5	9 7 5 3 1	-1, -1, -1, -1, -1	Strictly decreasing
4	6	4 4 5 2 2 7	5, 5, 7, 7, 7, -1	Duplicates present

Sample Run

```
Enter the size of the array: 5
Enter 5 elements: 4 5 2 10 8
Next greater elements: 5, 10, 10, -1, -1
```

Result

The program successfully uses pointers to pass the array to next_greater() and nested loops to locate the nearest greater element on the right of each array element. It correctly prints -1 when no greater element exists.